



# Bioreactor design for continuous dark fermentative hydrogen production

Kyung-Won Jung<sup>a</sup>, Dong-Hoon Kim<sup>b</sup>, Sang-Hyoun Kim<sup>c</sup>, Hang-Sik Shin<sup>a,\*</sup>

<sup>a</sup> Department of Civil and Environmental Engineering, KAIST, 373-1 Guseong-dong, Yuseong-gu, Daejeon 305-701, Republic of Korea

<sup>b</sup> Wastes Energy Research Center, Korea Institute of Energy Research, 102 Gajeong-ro, Yuseong-gu, Daejeon 305-343, Republic of Korea

<sup>c</sup> Department of Environmental engineering, Daegu University, Jillyang-eup, Gyeongsan-si, Gyeongbuk 712-714, Republic of Korea

## ARTICLE INFO

### Article history:

Received 30 November 2010

Received in revised form 16 March 2011

Accepted 17 March 2011

Available online 22 March 2011

### Keywords:

Dark fermentative H<sub>2</sub> production

Continuous systems

Suspended bioreactor

Immobilized bioreactor

Operational parameters

## ABSTRACT

Dark fermentative H<sub>2</sub> production (DFHP) has received increasing attention in recent years due to its high H<sub>2</sub> production rate (HPR) as well as the versatility of the substrates used in the process. For most studies in this field, batch reactors have been applied due to their simple operation and efficient control; however, continuous DFHP operation is necessary from economical and practical points of view. Continuous systems can be classified into two categories, suspended and immobilized bioreactors, according to the life forms of H<sub>2</sub> producing bacteria (HPB) used in the reactor. This paper reviews operational parameters for bioreactor design including pH, temperature, hydraulic retention time (HRT), and H<sub>2</sub> partial pressure. Also, in this review, various bioreactor configurations and performance parameters including H<sub>2</sub> yield (HY), HPR, and specific H<sub>2</sub> production rate (SHPR) are evaluated and presented.

© 2011 Elsevier Ltd. All rights reserved.

## 1. Introduction

Fossil fuels have played a fundamental role in industrial development and are responsible for fulfilling 80% of energy demand globally. However, this current energy system is now facing two fundamental problems: gradual depletion and environmental pollution. This lack of sustainability has led to extensive research on new alternative energy sources (Brey et al., 2006). Among various alternative energy sources, H<sub>2</sub> is regarded as the most promising future energy carrier, since it produces only water when combusted, generating a 2.75 times higher energy yield (122 kJ/g) than hydrocarbon fuels. Furthermore, H<sub>2</sub> fuel cells and related H<sub>2</sub> technologies provide an essential link between renewable energy sources and sustainable energy services (Levin et al., 2004).

H<sub>2</sub> production methods can be broadly divided into physico-chemical and biological processes. Recently, more than 90% of H<sub>2</sub> production has been achieved via steam reforming of hydrocarbons and coal gasification, due to the low costs involved in these processes. However, these methods have been criticized by the public and specialists, because they entail the use of fossil fuels, thus emitting a significant amount of greenhouse gases (Ewan and Allen, 2005). Given these perspectives, biological H<sub>2</sub> production assumes paramount importance as an alternative energy resource. Despite relatively lower yields of H<sub>2</sub> compared with photo-driven processes, dark fermentative H<sub>2</sub> production (DFHP)

is a promising method due to its higher rate of H<sub>2</sub> evolution in the absence of light sources and the transformation of waste into environmentally sound materials.

Therefore, this article presents an up-to date overview of current knowledge on important operation parameters and various reactor configurations in DFHP.

## 2. Reactor design parameters

The operational parameters such as inoculum preparation and start-up, pH, temperature, HRT, substrate concentration and liquid product inhibition, feedstock, H<sub>2</sub> partial pressure and nutrients are considered to have significant effects on the performance of DFHP. Defining their optimal ranges would provide important information in determining reactor and system size, materials, additional equipment, chemical reagents, and so on.

### 2.1. Inoculum preparation and start-up

It is more practical to use mixed cultures than pure cultures in engineering point of view, because they are simpler to operate and easier to control, and may have broader choice of feedstock (Valdez-Vazquez et al., 2005). Thus, numerous methods have been made to obtain H<sub>2</sub> producing inocula from various seeding sources such as anaerobic digester sludge, sewage sludge, compost, manure and soil. Physico-chemical attack was generally used to get the inocula on the basis that main H<sub>2</sub> producing bacteria, *Clostridium* sp., are spore-forming bacteria (Li and Fang, 2007).

\* Corresponding author. Tel.: +82 42 350 3613; fax: +82 42 350 8640.

E-mail address: [hs48@kaist.ac.kr](mailto:hs48@kaist.ac.kr) (H.-S. Shin).

At first, toxic chemical addition was applied to inhibit  $H_2$ -consuming reaction, especially killing methanogens. 2-Bromoehtanesulfonate (BES), acetylene and chloroform were added to anaerobic digester sludge and showed successful performances (Liang et al., 2002). However, it is seldom used today due to the high cost of chemicals and their ineffectiveness on non  $H_2$ -producing bacteria such as lactic and propionic acid bacteria.

Acid/base treatment was also effective in obtaining  $H_2$  producing inocula because microbial activity is hindered at low and high pH. Acid (pH 3) and base (pH 10) enrichment of sewage sludge increased the  $H_2$ -production potential by 200 and 333 times compared to control, respectively (Chen et al., 2002). Similarly, successful continuous operation using acid (pH 3–4) enriched microflora has been reported (Lin and Chou, 2004).

Heat treatment of seed sludge has mostly been used for screening of  $H_2$  producing bacteria. It is reported that heat treatment not only reduces the non-spore-forming bacteria but also activates clostridia spores to commence germination by altering the germination receptor (Hawkes et al., 2007). The heating temperature and time was varied from 75 °C to 121 °C and 15 min to 12 h, respectively (Li and Fang, 2007). Sometimes sequential pretreatment (heat shock at 100 °C for 2 h and acid treatment at pH 3.0 for 24 h) was applied for the perfect extermination of non  $H_2$ -producing bacteria (HPB) in the seeding source (Mohan et al., 2007). In addition to above main pretreatment methods, aeration was applied to compost sludge (Ueno et al., 1996).

After obtaining mixed cultures of  $H_2$  producing bacteria by various pretreatments, continuous operation has to be postponed until  $H_2$  production from batch operation reaches some extent in order to germinate  $H_2$  producing bacteria and to block their wash-out. Continuous operation was preceded when the  $H_2$  yield (HY) reached 0.5 mol  $H_2$ /mol hexose or after 48 h by batch mode (Kim et al., 2006a,b). Interestingly, Kim et al. (2008) observed the successful start-up only when the continuous feeding started after HY reached 0.2 mol  $H_2$ /mol hexose by batch mode at 12 h of HRT using continuously stirred tank reactor (CSTR).  $H_2$  production ceased within 10 d when the operation mode was changed from batch to continuous after 0.5 mol  $H_2$ /mol hexose of HY was achieved by batch mode and concluded that it was due to the regrowth of propionic acid bacteria which were inhibited by heat shock (90 °C for 20 min) but not totally exterminated (Kim et al., 2008). Generally, HRT decreased gradually for the successful acclimation of  $H_2$  producing bacteria in the reactor. It decreased from 5 d to 12 h in 4 steps, each taking 20 d (Lin and Jo, 2003), 120 to 12 h (Lin and Chou, 2004).

## 2.2. pH

Among various operational parameters, it has been widely accepted that pH has the most significant effect on DFHP, since it directly affects the hydrogenase activity, metabolic pathway, and dominant species (Lay, 2000; Fang et al., 2002). However, the optimal initial and operational pH values vary extensively from 4.5 to 9.0.

Some studies have shown that pH lower than 5.0 is preferable for  $H_2$  production.  $H_2$ -consuming methanogenic activity has been detected at pH 5.0 in studies by Kim et al. (2004) and Hwang et al. (2004). They concluded that a weakly acidic condition around 5.0 is not sufficient to exterminate methanogens, and therefore pH should be lowered to 4.5 so as to prohibit  $H_2$ -consuming reactions. Based on equilibrium of the NADH/NAD<sup>+</sup> ratio inside the cell, it was proposed that an acetate–ethanol fermentation type induced at pH 4.5 is a better and more stable metabolic pathway than acetate–butyrate or acetate–propionate metabolic pathways induced at pH between 5.0 and 7.5 (Ren et al., 1997).

Several studies concluded that the proper pH range is above 7.0. The maximum  $H_2$  production was detected at initial pH values of 9.0 and 7.5 by Lee et al. (2002) and Wang et al. (2006), respectively, using glucose as a substrate. Also, a maximum HY of 68.1 mL  $H_2$ /g TVS was observed at an initial pH 7.0 in treating wheat straw waste (Fan et al., 2006b). However, in the three aforementioned studies on the effects of initial pH, the pH was not controlled during fermentation; it was allowed to drop without any buffer addition. This experimental condition could lead to wrong conclusions, as the pH change is highly dependent on the substrate concentration and the amount of buffer capacity. For example, if there is an insufficient amount of buffer in the medium in treating a high-strength substrate, the pH drop will be drastic, and hence it is important to find the optimum pH at relatively higher ranges. However, if sufficient buffer is provided to this substrate and the buffer is also diluted, then a neutral initial pH also could yield high  $H_2$  production. Thus, in a batch process, it is necessary to separately define the roles of initial and operational pH and find the optimal values with consideration of the substrate concentration and buffer capacity.

The main anaerobic HPB, *Clostridium* sp., have several metabolic pathways, and hydrogenic reactions are dominant at pH 5.0–6.5, while non-hydrogenic reactions are triggered outside of this range (Jones and Woods, 1986). Therefore, the pH of recent  $H_2$  producing reactors is generally controlled at pH 5.0–6.5. In batch studies reported by Lay (2000) and Fan et al. (2006a), symmetric graphs were shown with pH 5.2 and 6.0 at the center peak, respectively. The alcohol production rate was greater than the  $H_2$  production rate (HPR) if the pH was lower than 4.3 or higher than 6.1 (Lay, 2000). A number of reports on continuous operation studies have also revealed that pH around 5.5 is optimal. Fang and Liu (2002) and Yu et al. (2002), concluded that the optimal pH was 5.5 in treating glucose and rice winery wastewater, respectively. Increased microbial diversity was observed at high pH (Fang and Liu, 2002), and the compositions of propionate and ethanol were increased at lower pH (Yu et al., 2002).

The addition of an alkaline solution to control pH is essential in DFHP, but presents an economic burden. To date, to our knowledge, the required amount of alkaline solution has never been quantified and an efficient way to reduce the amount has not been reported. Kraemer and Bagley (2005) attempted to decrease the alkaline solution requirement by recycling the effluent in a methane fermenter; however, this resulted in 87% decrease of  $H_2$  production, resulting from  $H_2$ -consuming methanogenic activity. We speculate that anaerobic co-digestion with a high buffer containing feedstock such as sewage sludge or livestock waste can be an economical solution. As these wastes are protein rich, a large amount of hydroxide ions along with ammonia ions (NH<sub>4</sub><sup>+</sup>) would be supplied during the fermentation, which would help to mitigate pH drop.

## 2.3. Temperature

Temperature affects the activity of microorganisms and the conversion rate of fermentation products, and is closely related to economic benefit. Zhang and Shen (2006) obtained results indicating that the sensitivity of mixed bacteria to temperature was significantly high and the optimal temperature was found to be around 35 °C. Mu et al. (2006a,b) examined the effect of temperature by varying the temperature from 33 to 41 °C and found that  $H_2$  production and microbial growth rate were increased with increased temperature, accompanied by a change of the metabolic product distribution.

In spite of placing an economic burden,  $H_2$  fermenters are often operated in a thermophilic (50–60 °C) or hyper-thermophilic (70–80 °C) range, since it is believed that operation at high

temperature could enhance the  $H_2$  fermentation performance by promoting hydrolysis and simplifying microbial diversity such that it is favorable to  $H_2$  production. Gavala et al. (2006) and Yu et al. (2002) observed enhanced  $H_2$  production from glucose and rice winery wastewater, respectively, by increasing the operation temperature from 35 to 55 °C. According to Shin et al. (2004) and Idania et al. (2005),  $H_2$  production obtained from a thermophilic culture was much higher than that of a mesophilic culture owing to changes in the population dynamics and metabolic pathways that are favorable to  $H_2$  production. The highest HY (2.47 mol  $H_2$ /mol glucose) in continuous operation reported to date was achieved under a hyper-thermophilic (70 °C) condition (Kotsopoulos et al., 2006), and complete disappearance of  $H_2$ -consuming activity was observed at 75 °C in treating cow waste slurry (Yokoyama et al., 2007). High temperature can promote hydrolysis and simplify microbial diversity in a manner favorable to  $H_2$  production, but it can also bring about monotonous microbial diversity, resulting in incomplete substrate degradation, especially in the treatment of actual waste. Also, operation at high temperature places an economic burden, as it requires a tight and closed structure and immense energy to heat and maintain the temperature of the reactor. Therefore, the temperature effect must be thoroughly investigated considering not only the  $H_2$  fermentation performance but also substrate degradation and economic factors.

#### 2.4. Hydraulic retention time (HRT)

As HRT is related to the amount of organics that can be handled per unit time, it has a direct impact on economical operation. Generally,  $H_2$  producing bacteria prefer short retention time, since the main  $H_2$  producing bacteria, *Clostridium* sp., tend to produce VFAs with  $H_2$  at the exponential growth phase while they produce alcohols at the stationary growth phase (Jones and Woods, 1986). With respect to microbial community, short HRT is also preferred. The washout of propionic acid bacteria, which consume  $H_2$  during their metabolism, was observed upon transition of the HRT from 8 to 6 h (Zhang et al., 2006). It is generally held that short HRT prohibits methanogenic growth, since the growth rate of methanogens is much lower than that of  $H_2$ -producing bacteria.

In the first (to the best of the authors' knowledge) review paper mainly focused on DFHP, it was concluded that an HRT in a range of 8–12 h is optimal for a liquid-type substrate (Hawkes et al., 2002). However, recently, the employment of immobilization and granulation techniques has dramatically reduced the HRT with increased HPR. This is discussed in detail in Section 3.2. In treating solid-type feedstock, the optimal HRT was much longer, as a hydrolysis step is required. Shin and Youn (2005) compared the  $H_2$  fermentation performance of food waste at HRTs of 2, 3, and 5 d; a HRT of 5 d showed the highest HY, 2.2 mol  $H_2$ /mol hexose. A survey of the literature indicates that the shortest HRT with stable and high performance achieved to date is 1.2 d, for the treatment of a mixture of food waste and paper waste operated under thermophilic (60 °C) conditions using packed-bed reactor (Ueno et al., 2007).

#### 2.5. Substrate concentration and liquid product inhibition

In the batch tests, optimal substrate concentration was quite varied and deeply influenced by other operational parameters such as pH. When the pH was not controlled, HY usually decreased with increasing substrate concentration due to low pH condition. Using fractional factorial design, the effects of pH and substrate concentration and their interaction on  $H_2$  production were investigated, and pH 5.5 at substrate concentration of 7.5 g COD/L was found to be the optimal condition (Ginkel et al., 2001).

Finding the optimal substrate concentration in continuous operation is more meaningful and practicable especially in treating

wastewater. Best performance was found at 30 g sucrose COD/L with HY of 1.09 mol  $H_2$ /mol hexose using CSTR (Kim et al., 2006b). At inlet substrate concentration below 20 g COD/L, the HY decreased along with a significant decrease in the *n*-butyrate/acetate ratio and appearance of  $H_2$  consuming bacteria and decrease of substrate removal efficiency was observed at over 35 g COD/L. However, the HY decreased from 2.4 to 1.7 mol  $H_2$ /mol hexose with substrate concentration increase from 2.5 to 10 g COD/L (Ginkel and Logan, 2005).

High substrate concentration allows more energy-efficient operation but product inhibition is likely to set the upper limit. Certain level of metabolic products in the DFHP may inhibit  $H_2$  producing pathway as well as microbial activity. It is known that butyrate has the most toxic effect on *Clostridium* sp., among various acids, thus a lot of attempts were made to alleviate butyrate inhibition, mostly by chemical extraction. Kyazze et al. (2005) reported that adding butyrate to a stable reactor to raise the free butyrate level from 33 to 63 mM stopped gas production completely, although the reactor recovered in 84 h after the spike.

#### 2.6. Feedstock

The main substrate for the fermentative  $H_2$  production was synthetic wastewater containing carbohydrate substances. Numerous works have been focused on glucose and sucrose and optimal process parameters including pH, HRT, temperature, substrate concentration, etc. for them were fully investigated (Li and Fang, 2007; Ghosh and Hallenbeck, 2010). Xylose and cellulose, main components of agricultural wastes, were also used for the successful  $H_2$  production. Polysaccharides such as starch were also successfully used as a single substrate for the  $H_2$  production (Arooj et al., 2007). Mohan et al. (2010) studied about the feasibility of fermentative effluents from DFHP as substrate for poly ( $\beta$ -OH) butyrate production. It was found that HY was significantly increased (12.48–16.21 mol  $H_2$ /kg COD) with increase in OLR (2.91–4.58 kg COD/m<sup>3</sup>/d).

For the dual benefit of fermentative  $H_2$  production, waste degradation and energy generation, actual wastes were often utilized as substrates. Carbohydrate-rich actual wastewaters were the main target for the  $H_2$  fermentation. Food processing wastewater (Ginkel and Logan, 2005), reed canary grass (RCG) (Lakaniemi et al., 2011), and rice straw (Lo et al., 2010) were proved to be feasible substrates by batch tests showing the maximum HY of 2.8 L  $H_2$ /L-wastewater, 1.25 mmol  $H_2$ /g RCG, and 0.76 mol  $H_2$ /mol xylose, respectively. Continuous  $H_2$  production from rice winery wastewater in an upflow reactor was conducted at various operating conditions and achieved the optimum HPR of 9.33 L  $H_2$ /g VSS/d at HRT of 2 h, COD concentration of 34 g COD/L, pH 5.5 and temperature 55 °C (Yu et al., 2002). High HY of 2.52 mol  $H_2$ /hexose was obtained using CSTR from sugary wastewater and it was successfully operated for 190 days (Ueno et al., 1996). Fan et al. (2006a) operated the CSTR treating brewery waste at various HRT and pH and achieved the maximum  $H_2$  production performance of 43 mL  $H_2$ /g COD at HRT = 18 h and pH = 5.5. Also cheese whey wastewater was tested for the  $H_2$  production using CSTR (Venetsaneas et al., 2009). Recently, Jung et al. (2011a,b) evaluated DFHP from eight different kinds of marine algae, often referred to as third generation biomass, in order to find new feedstock. Results showed that the highest HY of 69.1 mL  $H_2$ /g COD was obtained from *Laminaria japonica*. It was attributed to its high carbohydrate content (about 60%) in the cell, including laminarin and alginate.

#### 2.7. $H_2$ partial pressure

Partial pressure or  $H_2$  in the liquid phase is one of the key factors affecting DFHP. It is known that high  $H_2$  partial pressure

generally has a negative effect on  $H_2$  production, by decreasing the activity of hydrogenase and making the  $H_2$  production reaction thermodynamically unfavorable (Bahl and Dürre, 2001). Ruzicka (2006) theoretically proved that high  $H_2$  partial pressure inside the cell may facilitate propionate production while suppressing acetate and butyrate production. Numerous attempts have been made to decrease  $H_2$  partial pressure. Vigorous mixing to avoid supersaturation,  $N_2$  and  $CO_2$  sparging, and installation of a  $H_2$ -permeable membrane to remove dissolved  $H_2$  from the mixed liquor were found to be effective means of enhancing  $H_2$  production (Mizuno et al., 2000; Liang et al., 2002; Kim et al., 2006a).

## 2.8. Nutrients

Besides main substrate, carbohydrate materials, DFHP requires nutrients for bacterial activity like all biological treatment processes. The nutrients include nitrogen (N), phosphorous (P), ferrous (Fe) and some trace metals. Among many kinds of nutrients N is the most essential one for bacterial growth. N was supplied by various forms i.e. ammonia ion, yeast extract, peptone and so on. Optimal C/N ratio was varied a lot, 6.7 and 47 according to Liu and Shen (2004) and Lin and Lay (2004a), respectively. Excess ammonia addition caused inhibitory effect on  $H_2$  production; continuous decrease of HY from 1.9 to 1.1 mol  $H_2$ /mol hexose with increase of ammonia concentration from 0.8 to 7.8 g  $NH_4$ -N/L (Salerno et al., 2006).

P and Fe concentrations would affect metabolic pathway of *Clostridium* sp., and  $H_2$  production potential decreased when their concentrations were limited. P is the main source of ATP formation and it has a lot of functions in enzyme linkage. Lin and Lay (2004b) found that phosphate acted as a better buffer source for  $H_2$  production than carbonate. Its addition enhanced  $H_2$  production by 1.9 times and decreased the lag period. Effect of iron has investigated a lot in DFHP since it is an essential component of hydrogenase, but its optimal concentration was varied. It was varied with culture temperature; 800, 200 and 25 mg  $FeSO_4$ /L when cultured at 25, 35 and 40 °C, respectively (Zhang and Shen, 2006).

Lin and Lay (2005) studied the requirement of eleven trace metals in  $H_2$  fermentation. Magnesium, sodium, zinc and iron were found to be the important trace metals with magnesium the most significant one.  $H_2$  production enhanced by 30% at optimal combined concentrations, 4.8 mg  $Mg^{2+}$ /L, 393 mg  $Na^+$ /L, 0.25 mg  $Zn^{2+}$ /L and 1 mg  $Fe^{2+}$ /L. Li and Fang (2007) suggested the  $C_{1.50}$  values, at which the bioactivity of the sludge was reduced to 50% of the control, for individual heavy metals; Cu 30 mg/L, Ni and Zn 1600 mg/L, Cr 3000 mg/L, Cd 3500 mg/L and Pb >5000 mg/L and they concluded that  $H_2$ -producing sludges have higher resistance than methanogenic sludges.

## 3. Bioreactor classification

Two types of reactors have been developed for DFHP: (1) suspended bioreactors – continuously stirred tank reactor (CSTR), anaerobic sequencing batch reactor (ASBR), and anaerobic membrane bioreactor (AnMBR); and (2) immobilized bioreactors – up-flow anaerobic sludge blanket reactor (UASBr), expanded granular sludge bed reactor (EGSBr), and anaerobic fluidized bed reactor (AFBR).

### 3.1. Suspended bioreactors

Early applications of hydrogen fermentation utilized suspended bioreactors, which were initially designed to function in a manner similar to anaerobic digestion for methane fermentation. Although immobilized bioreactors have recently been developed, the sus-

pended bioreactors are still useful, especially for feedstock containing high particular content such as municipal solid waste and food waste. As highly concentrated biomass is a means of guaranteeing positive energy gain for hydrogen production (Hawkes et al., 2002), suspended bioreactors hold a dominant position in hydrogen fermentation. Table 1 lists feedstock, operating conditions, and  $H_2$  production performances for various bioreactor configurations.

#### 3.1.1. Continuously stirred tank reactor (CSTR)

The simplest  $H_2$ -producing suspended bioreactor configuration is a continuously stirred tank reactor (CSTR). The important operating parameters such as pH, substrate concentration, solids retention time (SRT), and hydrogen content have been derived mostly from CSTR operation (Hawkes et al., 2007). In a strict definition of a CSTR, feeding and discharge are also continuously performed; however, they are sometimes also performed in sequencing mode, especially for biomass with high particular content (Li et al., 2008). Table 2 summarizes the maximum  $H_2$  production in CSTRs fed with organic solid waste.

In most cases,  $H_2$ -producing suspended bioreactors have cylindrical shapes and employ mechanical turbines for mixing. As the size of a  $H_2$ -producing CSTR is much smaller than that of a conventional methane digester due to the shorter HRT of the former, the aforementioned shape and mixing method could be feasible up to a certain level of commercialization. However, it might be necessary to apply enhanced shapes and mixing methods to achieve ideal CSTR condition at larger sizes.

It has been recently reported that  $H_2$  producing bacteria (HPB) in CSTR operation can be rapidly flocculated and granulated. Although, few studies on the commencement of granulation and formation mechanisms of  $H_2$ -producing granule (HPG) have been reported, the formation of HPG can be explained by a four-step model (Schmidt and Ahring, 1996): (1) transportation of cells to the surface of other cells; (2) initial reversible adsorption to the substratum; (3) irreversible adhesion of the cells; (4) multiplication of the cells and formation of granules. It is also believed that divalent cations, e.g.,  $Ca^{2+}/Mg^{2+}/Fe^{2+}$ , and extracellular polymeric substance (EPS), especially the carbohydrate component of EPS, might play essential roles in the formation of HPG (Fang et al., 2002; Zhang et al., 2007). Fang et al. (2002) first reported that HPB could be agglutinated into granules at 26 °C, pH 5.5, and HRT 6 h within 80 days, having a diameter and settling velocity of 1.6 mm and 50 m/h, respectively, and consequently the HPR was 13.0 L  $H_2$ /L/d. Furthermore, the HPG contained EPS of 179 mg/g VSS and had a protein/carbohydrate (P/C) ratio of 0.2. Similarly, Zhang et al. (2007) reported that HPG was successfully formed after 114 h and consequently a HPR of 3.2 L  $H_2$ /L/h was achieved by acid incubation of sludge for 24 h. These findings suggest that an acidic environment will neutralize the negative charge of the cell surface, thereby decreasing the repulsion forces between cells. Initially, the portion of protein was higher than the amount of carbohydrate with a P/C ratio of 0.8; however, a decrease of the P/C ratio was observed and it was maintained at 0.2. Jung et al. (2011a,b) also demonstrated that the key EPS component for successful formation of HPG is carbohydrates. Indeed, it was reported that CDMW is a suitable feedstock to form HPG, because it has various divalent cations, especially calcium ions.

#### 3.1.2. Anaerobic sequencing batch reactor (ASBR)

Decoupling of SRT from HRT would provide a superior organic loading rate (OLR) and  $H_2$  productivity, because shorter HRT and higher retention of  $H_2$ -producers would be achieved.  $H_2$ -producing high-rate bioreactors initially followed the example of high-rate digesters for methane production. However, longer SRT might cause growth of  $H_2$ -consumers or competitors for the substrate, such as non-hydrogen producing acidogens (Chang et al., 2002).

**Table 1**Summary of feedstock, operating conditions, and H<sub>2</sub> production performances for various bioreactor configurations.

| Reactor type | Feedstock              | HRT (h) | pH        | Temperature (°C) | Maximum H <sub>2</sub> production   |                             | Reference                 |
|--------------|------------------------|---------|-----------|------------------|-------------------------------------|-----------------------------|---------------------------|
|              |                        |         |           |                  | HY (mol H <sub>2</sub> /mol hexose) | HPR (L H <sub>2</sub> /L/h) |                           |
| CSTR         | Food waste             | 42–24   | Above 5.3 | –                | 25.8 <sup>a</sup>                   | 0.04                        | Kim et al. (2008)         |
|              | Glucose                | 6       | 7–4       | 36               | 2.24                                | 0.54                        | Fang et al. (2002)        |
|              | Glucose                | 12–0.5  | 5.5       | 37               | 1.93                                | 3.2                         | Zhang et al. (2007)       |
|              | CDMW <sup>b</sup>      | 12–6    | 5.5 ± 0.2 | 35 ± 1           | 0.32                                | 0.34                        | Jung et al. (2010)        |
|              | Sugary wastewater      | 3–0.5   | 6.8       | 60               | 2.52                                | 0.17                        | Ueno et al. (1996)        |
|              | Cheese whey wastewater | 24      | 5.2       | 35               | 0.83                                | 0.12                        | Venetsaneas et al. (2009) |
| ASBR         | Starch                 | 18      | 5.3       | 35               | 0.5                                 | –                           | Arooj et al. (2007)       |
|              | Sucrose                | 12–4    | 6.7 ± 0.1 | 35 ± 1           | 1.46                                | –                           | Lin and Jo (2003)         |
|              | Starch                 | 18      | 5.3       | 35               | 0.2                                 | –                           | Arooj et al. (2007)       |
|              | Food waste             | 42–24   | Above 5.3 | –                | 80.9 <sup>a</sup>                   | 0.11                        | Kim et al. (2008)         |
| AnMBR        | Glucose                | 3.3–10  | 5.5       | –                | –                                   | 0.32                        | Oh et al. (2004)          |
|              | Fructose               |         |           |                  | 1.36                                | 2.75                        |                           |
|              | Glucose                | 4–1     | 6.8–6.2   | 35               | 1.27                                | 1.48                        | Lee et al. (2007)         |
|              | Sucrose                |         |           |                  | 1.39                                | 2.07                        |                           |
|              | Glucose                | 9       | 5.5 ± 0.1 | 35 ± 0.5         | 1.71                                | –                           | Lee et al. (2009)         |
| UASBr        | Glucose                | 8       | 5.5 ± 0.1 | 23 ± 1           | 1.03                                | 0.17                        | Shen et al. (2009)        |
|              | Sucrose                | 24–4    | 6.7 ± 0.2 | 35 ± 1           | 0.75                                | 0.25                        | Chang and Lin (2004)      |
|              | Sucrose                | 13      | 6.3–3.4   | 39 ± 1           | 1.68                                | 0.145                       | Mu et al. (2006a,b)       |
|              | Sucrose                | 17      | 6–3.1     | 35 ± 1           | 1.33                                | 0.1                         | Wang et al. (2007)        |
|              | Sucrose                | 30–18   | 4.5–4.3   | 38               | 1.62                                | –                           | Mu and Yu, 2006           |
|              | Sucrose                | 30–2    | 4.4 ± 0.1 | 38               | 1.54                                | –                           | Yu and Mu (2006)          |
|              | Sucrose                | 4–0.25  | 6.7       | 35               | 1.70                                | 7.3                         | Lee et al. (2004)         |
|              | Glucose                | 5.3     | –         | –                | –                                   | 0.48                        | Hu and Chen (2007)        |
|              | Glucose                | 26.7–24 | –         | 70               | 2.47                                | –                           | Kotsopoulos et al. (2006) |
|              | Rice winery wastewater | 24–2    | 6–4.5     | 55–20            | 2.14                                | 0.16                        | Yu et al. (2002)          |
| EGSBr        | CDMW                   | 12–6    | –         | 35 ± 1           | 1.78                                | 2.76                        | Jung et al. (2011a,b)     |
|              | Molasses wastewater    | 6–1     | –         | –                | 1.95                                | 0.71                        | Guo et al. (2008a)        |
|              | Starch wastewater      | 24–4    | 3.95      | 30 ± 1           | –                                   | 0.07                        | Guo et al. (2008b)        |
|              | Fresh leachate         | 24–6    | 5.0–5.5   | 35 ± 1           | –                                   | 0.09                        | Liu et al. (in press)     |
| AFBR         | Sucrose                | 6–1     | –         | 35               | 1.5                                 | 0.93                        | Wu et al. (2003)          |
|              | Glucose                | 3–0.125 | 5.5 ± 0.2 | 37               | 1.7                                 | 7.6                         | Zhang et al. (2008)       |
|              | Glucose                | 4–0.5   | –         | 37               | 1.24                                | 2.36                        | Zhang et al. (2007)       |
|              | Glucose                | 8–1     | –         | 30 ± 1           | 2.66                                | 0.97                        | Amorim et al. (2009)      |
|              | Glucose                | 8–1     | –         | 30 ± 1           | 2.44                                | 1.28                        | Shida et al. (2009)       |
|              | Glucose                | 8–1     | –         | 30               | 2.76                                | 1.21                        | Barros et al. (2010)      |
|              | Glucose                | 20–10   | –         | 70               | 1.31                                | 0.125                       | Peintner et al. (2010)    |
|              | Sucrose                | 8.9–2.2 | –         | –                | 2.79                                | 2.27                        | Lin et al. (2006)         |

<sup>a</sup> mL H<sub>2</sub>/g VS.<sup>b</sup> Coffee drink manufacturing wastewater.**Table 2**Maximum H<sub>2</sub> production in CSTR fed with organic solid waste.

| Feedstock                                 | Feed/Draw    | HRT (h) | pH      | Temperature (°C) | Maximum H <sub>2</sub> production <sup>a</sup> |                                   | Reference            |
|-------------------------------------------|--------------|---------|---------|------------------|------------------------------------------------|-----------------------------------|----------------------|
|                                           |              |         |         |                  | Rate (L H <sub>2</sub> /L/d)                   | Yield (mL/g VS <sub>added</sub> ) |                      |
| Garbage amended with shredded paper waste | Continuous   | 28.8    | 5.8–6.0 | 60               | 5.40                                           | 46.3 <sup>b</sup>                 | Ueno et al. (2007)   |
| Bean curd manufacturing waste             | Continuous   | 6       | 5.5     | 35               | 1.20                                           | 24.0                              | Noike et al. (2005)  |
| Brewery waste                             | Continuous   | 18      | 5.5     | 37               | 2.68                                           | 37.9                              | Fan et al. (2006a,b) |
| Food waste                                | Intermittent | 120     | 5.5     | 55               | 1.00                                           | 125.0                             | Shin and Youn (2005) |
| Household solid waste                     | Intermittent | 48      | 4.8–5.2 | 37               | 1.61                                           | 43.0                              | Liu et al. (2006)    |
| Kitchen waste                             | Intermittent | 48–96   |         |                  | 1.34                                           | 44.8 <sup>b</sup>                 | Li et al. (2008)     |

<sup>a</sup> All data were corrected to standard temperature (0 °C) and pressure (760 mm Hg).<sup>b</sup> mL/g COD<sub>added</sub>.

Therefore, the operating parameters and some configurations of H<sub>2</sub>-producing high-rate bioreactors should be different from those for methane production.

The anaerobic sequencing batch reactor (ASBR) is a batch variation of up-flow reactors such as the UASBr in that it depends on the development of superior settling biomass and provides for staging of the kinetics (Sung and Dague, 1992). The process involved batch feeding, internal solids separation, and supernatant wasting, and is capable of achieving long SRT with relatively short HRT. It is suitable for treating feedstock containing high particular content, unlike other high-rate digestion processes (Dugba and Zhang, 1999). The simplicity of the operation by using batch feeding is another advantage in solid waste treatment (Han and Shin, 2004).

Lin and Jo (2003) investigated the ASBR performance for hydrogen production from sucrose. HRT and the reaction period/settling period (R/S) ratio varied from 16 to 8 h and 1.7 to 5.6, respectively. VSS concentration was maintained at more than 5.0 g/L, which was much higher than that (1.5–2.5 g/L) in CSTR (Hawkes et al., 2002). However, the optimum HRT was 12 h, similar to that achieved in CSTR. The maximum hydrogen production of 1.3 mol H<sub>2</sub>/mole hexose<sub>added</sub> was attained at a R/S ratio 5.6 in the study by Hawkes et al. (2002).

Arooj et al. (2007) operated an ASBR (HRT 18 h and SRT 72 h) and a CSTR (HRT 18 h) with starch as a substrate, and compared their performance. The ASBR showed higher VSS concentration, but lower H<sub>2</sub> production (0.2 mol H<sub>2</sub>/mol hexose in ASBR vs. 0.5 mol H<sub>2</sub>/mol hexose in CSTR). They found that non-settleable

decanting microorganisms, resulting from stratification in the settling phase, were capable of more specific  $H_2$  activity than settleable microorganisms in ASBR. On the other hand, Kim et al. (2008) reported higher hydrogen productivity of an ASBR (2.72 L  $H_2$ /L/d and 80.9 mL  $H_2$ /g VS at HRT 30 h and SRT 90 h) compared to that of a CSTR (0.91 L  $H_2$ /L/d and 25.8 mL  $H_2$ /g VS at HRT 30 h) with food waste as a substrate. They decanted the liquid from the middle or bottom layer (0.20 to 0.42 of decanting level, which is the relative height of the drawing port with respect to the depth of mixed liquor).

### 3.1.3. Anaerobic membrane bioreactor (AnMBR)

The membrane bioreactor (MBR) has emerged as an effective means of attaining performance improvement in wastewater treatment and has recently been applied to anaerobic process due to its capability of increasing biomass retention via membrane separation (Lee et al., 2010; Vallero et al., 2005). There are two principal approaches for membrane design and operation: external cross-flow type and submerged or immersed type. Previously, most researches have utilized an external cross-flow type due to ease of use and possible construction of a CSTR. However, the submerged type has attracted more interest recently and seen increased use, because it is less energy intensive and requires less membrane management (Yang et al., 2006; Lee et al., 2009).

Attempts have been made recently to apply the MBR process to hydrogen production, but relatively little research has been carried out thus far. Oh et al. (2004) demonstrated that HPR was increased by 25% to 0.32 L/L/h due to a 164% increase in biomass concentration to 5.8 g/L with an increase of the SRT from 3.3 to 12 h using an external cross-flow membrane. However, increasing the SRT from 5 to 48 h at a HRT of 5 h more increased overall glucose removal, but did not increase overall  $H_2$  production compared to that achieved with a HRT = SRT condition. Similarly, stable operation was observed in treating different kinds of carbon sources, fructose (2.75 L/L/h), glucose (1.48 L/L/h), and sucrose (2.07 L/L/h), by effectively avoiding cell washout, at 1 h HRT using a hollow-fiber membrane installation. However, decreased substrate utilization was observed, thereby decreasing HRT conditions (Lee et al., 2007). Lee et al. (2009) also reported that the HPR of the MBR was around 2.6 times higher than that of a CSTR, whereas no parallel increase between HPR and HY was found in the MBR. Indeed, due to a low F/M ratio of 2.6, the specific  $H_2$  production rate (SHPR) was considerably lower than that in the CSTR, specifically by about 15 times. Accordingly, previous reports pointed out that SRT and HRT are important factors in cross-flow and submerged membrane type for  $H_2$  production using an AnMBR, because SRT and HRT are closely related with the F/M ratio and organic loading rate (OLR), respectively. Lee et al. (2010) showed that increased  $H_2$  production was observed with increasing SRT, but it decreased at a long 90 d SRT. This might have been due to the low VSS/TSS concentration and a metabolic pathway shift to lactate. Moreover, decreased  $H_2$  production under a long SRT was due to a reduction of microbial growth, linked to the accumulation of EPS. Several reports have demonstrated that EPS can negatively affect not only microbial activity by metal toxicity but also membrane permeability by membrane fouling (Li and Fang, 2007; Lee et al., 2008; Halbouni et al., 2008; Shen et al., 2010). Membrane fouling is a key process limitation and remains one of the most challenging issues in future MBR development (Yang et al., 2006). However, the mechanisms of membrane fouling have not been clearly identified thus far. Two reports have attempted to define the mechanism of membrane fouling during DFHP (Lee et al., 2008; Shen et al., 2010). Both researches reported that colloidal adhesion and biomass deposition by EPS production directly led to a decrease in the permeate flux, and thus these were postulated as the main mechanisms of membrane fouling.

## 3.2. Immobilized bioreactors

Most outstanding  $H_2$  productivities reported to date have been achieved using immobilized reactors. The common reactor configurations are the UASBr, EGSBr, and AFBR.

### 3.2.1. Up-flow anaerobic sludge blanket reactor (UASBr)

The concept of the UASBr was developed in the 1970s for  $CH_4$  production, and it has prevailed successfully for anaerobic treatment of various types of wastewater to produce methane, because of its high treatment efficiency and excellent process stability (Lettinga et al., 1980; Wang et al., 2007). This reactor has a longitudinal structure with a gas/liquid/solid separator at the top, where microbial granules with high settling velocity are formed, resulting in a thick biomass blanket zone at the bottom (Jung et al., 2010).

Numerous works have dealt with  $H_2$  producing UASBr, since HPG formation was first reported by Fang et al. (2002), and as mentioned above, this reactor generally shows high and stable performance. However, for most studies in this field, synthetic wastewater is generally applied as a substrate. Chang and Lin (2004) produced  $H_2$  from sucrose using an UASBr seeded with heat pretreated sewage sludge. The highest HY and HPR values were 0.75 mol  $H_2$ /mol hexose and 0.25 L  $H_2$ /L/h, respectively, at a HRT of 8 h. In addition, the average HPG diameter was 0.43 mm after 173 d of operation. Mu et al. (2006a,b) attempted to optimize the operational pH condition from sucrose using an UASBr seeded with sludge taken from an anaerobic reactor treating citrate-producing wastewater. A maximum HY of 1.68 mol  $H_2$ /mol hexose and a HPR of 0.145 L  $H_2$ /L/h were obtained at pH 4.2. However, a long start-up period of 300 d was required for successful formation of HPG. Similarly, the highest HY and HPR values of 3.0 mol  $H_2$ /mol hexose and 0.05 L  $H_2$ /L/h, respectively, were achieved at 18 h HRT after a 150 day start-up period. Wang et al. (2007) reported the highest HY of 1.33 mol  $H_2$ /mol hexose and a HPR of 0.1 L  $H_2$ /L/h using sucrose as a substrate; however, a start-up period of 150 d was needed to establish stable operation. Lee et al. (2004) developed a carrier-induced granular sludge bed (CIGSB) reactor and installed carrier matrices on the bottom to stimulate granule formation in consideration of the findings of a previous study showing that self-flocculated sludge formed when the bed porosity was high (>90%) and the HRT was low (<4 h) (Lee et al., 2003). They achieved a maximum HPR of 7.3 L  $H_2$ /L/h at a 0.5 h HRT. Similarly, Wu et al. (2005) designed a reactor containing silicone-immobilized and self-flocculated sludge and obtained the highest HPR (15 L  $H_2$ /L/h) documented to date. A high concentration of biomass up to 35.4 g VSS/L was maintained even at 0.5 h of HRT. However, application of the UASBr is obstructed by the major drawback of a long start-up period, where a few months are generally required for HPG formation, as mentioned above.

Hu and Chen (2007) used methanogenic granules as a seeding source after heat, acid or chemical shock and observed whether this granular structure was maintained with  $H_2$  production ability. Only chemical shock (chloroform addition) caused irreversible damage to methanogens while  $H_2$  productivity reached 11.6 L  $H_2$ /L/d without granular structure breakage. In an effort to decrease the start-up period in the UASBr, Jung et al. (2011a,b) inoculated heat-treated sludge to a CSTR, and then mixed liquor in the CSTR was transferred to the UASBr as a seeding source. As a result, HPG with an average size of 1.9 mm was successfully formed in the UASBr after 45 d of operation using coffee drink manufacturing wastewater (CDMW). This is the first report on the formation of HPG from actual wastewater.

Thus far, the application of actual wastewater for DFHP using an UASBr has been limited, although it has high potential for significant enhancement of economic viability. Four studies on DFHP

from actual wastewater have been conducted thus far (Yu et al., 2002; Huang et al., 2004; Jung et al., 2010, 2011a,b). Rice winery wastewater was applied as a substrate for DFHP in order to optimize HRT, substrate concentration, and operational pH conditions (Yu et al., 2002). A maximum HY of 2.14 mol H<sub>2</sub>/mol hexose and a HPR of 0.16 L H<sub>2</sub>/L/h were reported; however, no information on the HPG was provided. Jung et al. (2011a,b) reported that maximum HY and HPR average values of 1.78 mol H<sub>2</sub>/mol hexose and 2.76 L H<sub>2</sub>/L/h, respectively, were obtained from CDMW.

### 3.2.2. Expanded granular sludge bed reactor (EGSBr)

An expanded granular sludge bed reactor (EGSBr) is a variant of the UASBr concept, and was developed to overcome problems such as preferential flows, hydraulic short cuts, and dead zones that can occur in the UASBr (Kato et al., 2003). In this reactor concept, a liquid up-flow velocity ( $V_{up}$ ) above 4 m/h permits partial expansion (fluidization) of the granular sludge bed, resulting in better substrate-sludge contact as well as enhancing separation of small inactive suspended particles from the sludge bed (Seghezzo et al., 1998). Due to the hydrodynamic characteristics in the EGSBr, soluble pollutants are efficiently treated but wastewater containing inert or poorly biodegradable suspended solids must not be allowed to accumulate in the reactor (Kato et al., 2003; Seghezzo et al., 1998). Recently, a few studies have been conducted to enhance DFHP using EGSBr; however, HPR was not substantially increased compared to the UASBr. Guo et al. (2008a) operated an EGSBr with granular activated carbon as the support media and achieved a highest HPR of 17.04 L H<sub>2</sub>/L/d and a HY of 1.95 mol H<sub>2</sub>/mol hexose under an OLR of 120 g COD/L/d (2 h HRT and 10 g COD/L) using molasses wastewater as a substrate. Guo et al. (2008b) also reported that the maximum HPR was 1.64 L H<sub>2</sub>/L/d under an OLR of 1.0 g starch/L/d and a HRT of 4 h using starch wastewater. Fresh leachate originating from municipal solid wastes was utilized for the first time as a substrate in an EGSBr (Liu et al., 2010). Results showed that the average HPR was 2.15 L H<sub>2</sub>/L/d under a  $V_{up}$  of 3.7 m/h and an OLR of 24 g COD/L/d.

### 3.2.3. Anaerobic fluidized-bed reactor (AFBR)

The AFBR system has been widely applied for wastewater treatment due to its high efficiency at low HRT with high biomass concentration (Hickey and Owens, 1981); accordingly, this system has been recently utilized for DFHP. In most studies in this field, gel entrapping, granular active carbon (GAC), and expanded clay were applied in order to immobilize HPB. The alginate gel entrapment method, i.e., mixing biomass with acrylic latex/silicon supplemented by a small amount of sodium alginate and activated carbon, was applied to a fluidized-bed reactor, resulting in a high HPR of 0.93 L H<sub>2</sub>/L/h at a short HRT of 2 h (Wu et al., 2003). Lin et al. (2006) operated a draft tube AFBR containing silicone gel immobilized HPB, and the highest HPR of 2.27 L H<sub>2</sub>/L/d was achieved under a HRT of 2.2 h. However, it was reported that the immobilized cells created by gel entrapping techniques easily suffer from mass transfer resistance and immobilized bioparticles can be damaged by the biogas produced inside the gel (Zhang et al., 2007).

Zhang et al. (2007) produced H<sub>2</sub> from glucose using GAC as a support medium. Results showed a maximum HPR of 2.36 L H<sub>2</sub>/L/h with an attached biofilm concentration of 21.5 g/L under a HRT of 1 h. Also, Zhang et al. (2008) reported that a biofilm-based AFBR using a GAC showed a higher HPR (7.6 L H<sub>2</sub>/L/h) as compared with a granule-based AFBR (6.6 L H<sub>2</sub>/L/h) under a HRT of 0.25 h with an average biomass concentration of 35 g VSS/L in both AFBR. In the case of using expanded clay as a support medium, Shida et al. (2009) and Amorim et al. (2009) reported that the highest HPR values were 1.28 L H<sub>2</sub>/L/h and 0.97 L H<sub>2</sub>/L/h when the HRT was 1 h, whereas the HYs were higher under a HRT of 2 h in both

studies. Barros et al. (2010) operated an AFBR using different support materials – polystyrene and expanded clay – and achieved a highest HPR of 1.21 L H<sub>2</sub>/L/h with 1.1 mg TVS/g using expanded clay under a HRT of 1 h.

## 4. Concluding remarks

Thus far, the numerous researches on DFHP have been investigated; however, the technology is still ambiguous to practically apply in full-scale operation. First, even though physico/chemical shock to obtain HPB is not a big economic burden in a lab-scale system, it could be a significant problem in full-scale system. Therefore, it is better to develop a practically applicable method, which is cheap but effective, or, it may be possible to obtain the inoculum from the substrate itself such as food waste (Kim et al., 2009).

Second, this review showed that immobilized bioreactors can overcome the drawbacks of suspended bioreactors, resulting in enhanced H<sub>2</sub> productivity. Nevertheless, numerous studies have focused on DFHP using carbohydrate-rich synthetic wastewater such as glucose and sucrose. Therefore, the utilization of actual waste/wastewater is strongly recommended in order to exploit the dual benefits of DFHP, waste degradation and energy generation. In addition, terrestrial biomass, referred to as first and second generation biomass has been a main substrate for biofuel. However, some negative opinions have been arisen due to several reasons (Jung et al., 2011a,b). Therefore, it is time to focus on aquatic biomass, such as micro and macro algae, due to its advantages as follows: (1) fast growth rate; (2) negligible lignin; (3) high CO<sub>2</sub> utilization rate.

Third, many operational parameters affect DFHP, and thus statistical approaches and kinetic models can help attain a better understanding of individual and interactive influences of each parameter and can provide useful information for the analysis, design, and operation of fermentation processes. Last, as the DFHP alone could convert, even at optimal condition, less than 33% of the substrate, second step process should be recommended. Recently, there has been an attempt to recover H<sub>2</sub>, CH<sub>4</sub>, and electricity by microbial fuel cell (MFC) from biodiesel or bioethanol waste via two-stage fermentation system. Although two-stage fermentation system would enhance total bioenergy conversion efficiency, it is still required further studies for practical application.

## Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MEST) (No. 2010-0026904)

## References

- Amorim, E.L.C., Barros, A.R., Damianovic, M.H.R.Z., Silva, E.L., 2009. Anaerobic fluidized bed reactor with expanded clay as support for hydrogen production through dark fermentation of glucose. *Int. J. Hydrogen Energy* 34, 783–790.
- Arooj, M.F., Han, S.K., Kim, S.H., Kim, D.H., Shin, H.S., 2007. Sludge characteristics in anaerobic SBR system producing hydrogen gas. *Water Res.* 41, 1177–1184.
- Bahl, H., Dürre, P., 2001. *Clostridia: Biotechnology and Medical Applications*. Wiley-VCH Verlag GmbH, Weinheim. pp. 57–63.
- Barros, A.R., Amorim, E.L.C., Reis, C.M., Shida, G.M., Silva, E.L., 2010. Biohydrogen production in anaerobic fluidized bed reactors: effect of support material and hydraulic retention time. *Int. J. Hydrogen Energy* 35, 3379–3388.
- Brey, J.J., Brey, R., Carazo, A.F., Contreras, I., Hernandez-Diaz, A.G., Castro, A., 2006. Designing a gradual transition to hydrogen economy in Spain. *Int. J. Hydrogen Energy* 31, 1231–1240.
- Chang, F.Y., Lin, C.Y., 2004. Biohydrogen production using an up-flow anaerobic sludge blanket reactor. *Int. J. Hydrogen Energy* 29, 33–39.
- Chang, J.S., Lee, K.S., Lin, P.J., 2002. Biohydrogen production with fixed-bed bioreactors. *Int. J. Hydrogen Energy* 27, 1167–1174.
- Chen, C.C., Lin, C.Y., Lin, M.C., 2002. Acid-base enrichment enhances anaerobic hydrogen production process. *Appl. Microbiol. Biotechnol.* 58, 224–228.

- Dugba, P.N., Zhang, R., 1999. Treatment of dairy wastewater with two-stage anaerobic sequencing batch reactor systems—thermophilic versus mesophilic operations. *Bioresour. Technol.* 68, 225–233.
- Ewan, B.C.R., Allen, R.W.K., 2005. A figure of merit assessment of the routes to hydrogen. *Int. J. Hydrogen Energy* 30, 809–819.
- Fan, K.S., Kan, N.R., Lay, J.J., 2006a. Effect of hydraulic retention time on anaerobic hydrogenesis in CSTR. *Bioresour. Technol.* 97, 84–89.
- Fan, Y.T., Zhang, G.S., Guo, X.Y., Xing, Y., Fan, M.H., 2006b. Biohydrogen-production from beer less biomass by cow dung compost. *Biomass Bioenerg.* 30, 493–496.
- Fang, H.H.P., Liu, H., Zhang, T., 2002. Characterization of a hydrogen-producing granular sludge. *Biotechnol. Bioeng.* 78, 44–52.
- Fang, H.H.P., Liu, H., 2002. Effect of pH on hydrogen production from glucose by a mixed culture. *Bioresour. Technol.* 82, 87–93.
- Gavala, H.N., Skiadas, I.V., Ahring, B.K., Lyberatos, G., 2006. Thermophilic anaerobic fermentation of olive pulp for hydrogen and methane production: modelling of the anaerobic digestion process. *Water Sci. Technol.* 53, 271–279.
- Ghosh, D., Hallenbeck, P.C., 2010. Response surface methodology for process parameter optimization of hydrogen yield by the metabolically engineered strain *Escherichia coli* DJT135. *Bioresour. Technol.* 101, 1820–1825.
- Ginkel, S.V., Logan, B., 2005. Increased biological hydrogen production with reduced organic loading. *Water Res.* 39, 3819–3826.
- Ginkel, S.V., Lay, J.J., Sung, S., 2001. Biohydrogen production as a function of pH and substrate concentration. *Environ. Sci. Technol.* 35, 4726–4730.
- Guo, W.Q., Ren, N.Q., Chen, Z.B., Liu, B.F., Wang, X.J., Xiang, W.S., Ding, J., 2008a. Simultaneous biohydrogen production and starch wastewater treatment in an acidogenic expanded granular sludge bed reactor by mixed culture for long-term operation. *Int. J. Hydrogen Energy* 33, 7397–7404.
- Guo, W.Q., Ren, N.Q., Wang, X.J., Xiang, W.S., Meng, Z.H., Ding, J., Qu, Y.Y., Zhang, L.S., 2008b. Biohydrogen production from ethanol-type fermentation of molasses in an expanded granular sludge bed (EGSB) reactor. *Int. J. Hydrogen Energy* 33, 4981–4988.
- Halbouni, D., Traber, J., Lyko, S., Wintgens, T., Melin, T., Tacke, D., 2008. Correlation of EPS content in activated sludge at different sludge retention times with membrane fouling phenomena. *Water Res.* 42, 1475–1488.
- Han, S.K., Shin, H.S., 2004. Biohydrogen production by anaerobic fermentation of food waste. *Int. J. Hydrogen Energy* 29, 569–577.
- Hawkes, F.R., Dinsdale, R., Hawkes, D.L., Hussy, I., 2002. Sustainable fermentative hydrogen production: challenges for process optimization. *Int. J. Hydrogen Energy* 27, 1339–1347.
- Hawkes, F.R., Hussy, I., Kyazze, G., Dinsdale, R., Hawkes, D.L., 2007. Continuous dark fermentative hydrogen production by mesophilic microflora: principles and progress. *Int. J. Hydrogen Energy* 32, 172–184.
- Hickey, R.F., Owens, R.W., 1981. Methane generation from high strength industrial wastes with the anaerobic biological fluidized bed. *Biotechnol. Bioeng. Symp.* 11399–11413.
- Hu, B., Chen, S., 2007. Pretreatment of methanogenic granules for immobilized hydrogen fermentation. *Int. J. Hydrogen Energy* 32, 3266–3273.
- Huang, G.H., Hsu, S.F., Liang, T.M., Huang, Y.H., 2004. Study on hydrogen production with hysteresis in UASB. *Chemosphere* 54, 815–821.
- Hwang, M.H., Jang, N.J., Hyun, S.H., Kim, I.S., 2004. Anaerobic bio-hydrogen production from ethanol fermentation: the role of pH. *J. Biotechnol.* 111, 297–309.
- Idania, V.V., Elvira, R.L., Fernando, E.G., Franco, C., Héctor, M.P.V., 2005. Semi-continuous solid substrate anaerobic reactors for H<sub>2</sub> production from organic waste: mesophilic versus thermophilic regime. *Int. J. Hydrogen Energy* 30, 1383–1391.
- Jones, D., Woods, D., 1986. Acetone–butanol fermentation revisited. *Microbiological Rev.* 50, 484–524.
- Jung, K.W., Kim, D.H., Shin, H.S., 2010. Continuous fermentative hydrogen production from coffee drink manufacturing wastewater by applying UASB reactor. *Int. J. Hydrogen Energy* 35, 13370–13378.
- Jung, K.W., Kim, D.H., Shin, H.S., 2011a. A simple method to reduce the start-up period in a H<sub>2</sub>-producing UASB reactor. *Int. J. Hydrogen Energy* 36, 1466–1473.
- Jung, K.W., Kim, D.H., Shin, H.S., 2011b. Fermentative hydrogen production from *Laminaria japonica* and optimization of thermal pretreatment conditions. *Bioresour. Technol.* 102, 2745–2750.
- Kato, M.T., Florencio, L., Arantes, R.F., 2003. Post-treatment of UASB effluent in an expanded granular sludge bed reactor type using flocculent sludge. *Water Sci. Technol.* 48, 279–284.
- Kim, D.H., Han, S.K., Kim, S.H., Shin, H.S., 2006a. Effect of gas sparging on continuous fermentative hydrogen production. *Int. J. Hydrogen Energy* 31, 2158–2169.
- Kim, D.H., Han, S.K., Shin, H.S., 2006b. Effect of substrate concentration on hydrogen production and 16S rDNA-based analysis of the microbial community in a continuous fermenter. *Process Biochem.* 41, 199–207.
- Kim, D.H., Kim, S.H., Ko, I.B., Lee, C.Y., Shin, H.S., 2008. Start-up strategy for continuous fermentative hydrogen production: early switchover from batch to continuous operation. *Int. J. Hydrogen Energy* 33, 1532–1541.
- Kim, D.H., Kim, S.H., Shin, H.S., 2009. Hydrogen fermentation of food waste without inoculum addition. *Enzyme Microbial Technol.* 45, 181–187.
- Kim, I.S., Hwang, M.H., Jang, N.J., Hyun, S.H., Lee, S.T., 2004. Effect of low pH on the activity of hydrogen utilizing methanogen in bio-hydrogen process. *Int. J. Hydrogen Energy* 29, 1133–1140.
- Kim, S.H., Han, S.K., Shin, H.S., 2008. Optimization of continuous hydrogen fermentation of food waste as a function of solids retention time independent of hydraulic retention time. *Process Biochem.* 43, 213–218.
- Kotsopoulos, T.A., Zeng, R.J., Angelidaki, I., 2006. Biohydrogen production in granular up-flow anaerobic sludge blanket (UASB) reactors with mixed cultures under hyper-thermophilic temperature (70°C). *Biotechnol. Bioeng.* 94, 296–301.
- Kraemer, J.T., Bagley, D.M., 2005. Continuous fermentative hydrogen production using a two-phase reactor system with recycle. *Environ. Sci. Technol.* 39, 3819–3825.
- Kyazze, G., Martinez-Perez, N., Dinsdale, R., Premier, G.C., Hawkes, F.R., Guwy, A.J., Hawkes, D.L., 2005. Influence of substrate concentration on the stability and yield of continuous biohydrogen production. *Biotechnol. Bioeng.* 93, 971–979.
- Lakaniemi, A.M., Koskinen, P.E.P., Nevatalo, L.M., Kaksonen, A.H., Puhakka, J.A., 2011. Biogenic hydrogen and methane production from reed canary grass. *Biomass and Bioenergy* 35, 773–780.
- Lay, J.J., 2000. Modeling and optimization of anaerobic digested sludge converting starch to hydrogen. *Biotechnol. Bioeng.* 68, 269–278.
- Lee, D.Y., Li, Y.Y., Noike, T., 2009. Continuous H<sub>2</sub> production by anaerobic mixed microflora in membrane bioreactor. *Bioresour. Technol.* 100, 690–695.
- Lee, D.Y., Li, Y.Y., Noike, T., 2010. Influence of solids retention time on continuous H<sub>2</sub> production using membrane bioreactor. *Int. J. Hydrogen Energy* 35, 52–60.
- Lee, D.Y., Li, Y.Y., Noike, T., Cha, G.C., 2008. Behavior of extracellular polymers and bio-fouling during hydrogen fermentation with a membrane bioreactor. *J. Membrane Sci.* 322, 13–18.
- Lee, K.S., Lin, P.J., Fangchiang, K., Chang, J.S., 2007. Continuous hydrogen production by anaerobic mixed microflora using a hollow-fiber microfiltration membrane bioreactor. *Int. J. Hydrogen Energy* 32, 950–957.
- Lee, K.S., Lo, Y.S., Lo, Y.C., Lin, P.J., Chang, J.S., 2003. H<sub>2</sub> production with anaerobic sludge using activated-carbon supported packed-bed bioreactors. *Biotechnol. Lett.* 25, 133–138.
- Lee, K.S., Wu, J.F., Lo, Y.S., Lo, Y.C., Lin, J., Chang, J.S., 2004. Anaerobic hydrogen production with an efficient carrier-induced granular sludge bed bioreactor. *Biotechnol. Bioeng.* 87, 648–657.
- Lee, Y.J., Miyahara, T., Noike, T., 2002. Effect of pH on microbial hydrogen fermentation. *J. Chem. Technol. Biotechnol.* 77, 694–698.
- Lettinga, G., Van Velsen, A.F.M., Hobma, S.W., De Zeeuw, W., Klapwijk, A., 1980. Use of upflow sludge blanket (USB) reactor concept for biological waste water treatment especially for anaerobic treatment. *Biotechnol. Bioeng.* 22, 699–734.
- Levin, D.B., Pitt, L., Love, M., 2004. Biohydrogen production: Prospects and limitations to practical application. *Int. J. Hydrogen Energy* 29, 173–185.
- Li, C.H., Fang, H.H.P., 2007. Fermentative hydrogen production from wastewater and solid wastes by mixed cultures. *Crit. Rev. Env. Sci. Tec.* 37, 1–39.
- Li, S., Kuo, S.C., Lin, J.S., Lee, Z.K., Wang, Y.H., Cheng, S.S., 2008. Process performance evaluation of intermittent–continuous stirred tank reactor for anaerobic hydrogen fermentation with kitchen waste. *Int. J. Hydrogen Energy* 33, 1522–1531.
- Liang, T.M., Cheng, S.S., Wu, K.L., 2002. Behavioral study on hydrogen fermentation reactor installed with silicone rubber membrane. *Int. J. Hydrogen Energy* 27, 1157–1165.
- Lin, C.N., Wu, S.Y., Chang, J.S., 2006. Fermentative hydrogen production with a draft tube fluidized bed reactor containing silicone-gel-immobilized anaerobic sludge. *Int. J. Hydrogen Energy* 31, 2200–2210.
- Lin, C.Y., Chou, C.H., 2004. Anaerobic hydrogen production from sucrose using an acid-enriched sewage sludge microflora. *Eng. Life. Sci.* 4, 66–70.
- Lin, C.Y., Lay, C.H., 2005. A nutrient formulation for fermentative hydrogen production using anaerobic sewage sludge microflora. *Int. J. Hydrogen Energy* 30, 285–292.
- Lin, C.Y., Lay, C.H., 2004a. Carbon/nitrogen-ratio effect on fermentative hydrogen production by mixed microflora. *Int. J. Hydrogen Energy* 29, 41–45.
- Lin, C.Y., Lay, C.H., 2004b. Effects of carbonate and phosphate concentrations on hydrogen production using anaerobic sewage sludge microflora. *Int. J. Hydrogen Energy* 29, 275–281.
- Lin, C.Y., Jo, C.H., 2003. Hydrogen production from sucrose using an anaerobic sequencing batch reactor process. *J. Chem. Technol. Biotechnol.* 78, 678–684.
- Liu, D.W., Liu, D.P., Zeng, R.J., Angelidaki, I., 2006. Hydrogen and methane production from household solid waste in the two-stage fermentation process. *Water Res.* 40, 2230–2236.
- Liu, G., Shen, J., 2004. Effects of culture and medium conditions on hydrogen production from starch using anaerobic bacteria. *J. Biosci. Bioeng.* 98, 251–256.
- Liu, Q., Zhang, X., Yu, L., Zhao, A., Tai, J., Liu, J., Qian, J., Xu, Z.P., 2010. Fermentative hydrogen production from fresh leachate in batch and continuous bioreactors. *Bioresour. Technol.* In Press.
- Lo, Y.C., Lu, W.C., Chen, C.Y., Chang, J.S., 2010. Dark fermentative hydrogen production from enzymatic hydrolyzate of xylan and pretreated rice straw by *Clostridium butyricum* CGS5. *Bioresour. Technol.* 101, 5885–5891.
- Mizuno, O., Dinsdale, R., Hawkes, F.R., Hawkes, D.L., Noike, T., 2000. Enhancement of hydrogen production from glucose by nitrogen gas sparging. *Bioresour. Technol.* 73, 59–65.
- Mohan, S.V., Bhaskar, Y.V., Krishna, P.M., Rao, N.C., Babu, V.L., Sama, P.L., 2007. Biohydrogen production from chemical wastewater as substrate by selectively enriched anaerobic mixed consortia: Influence of fermentation pH and substrate composition. *Int. J. Hydrogen Energy* 32, 2286–2295.
- Mohan, S.V., Reddy, M.V., Subhash, G.V., Sarma, P.N., 2010. Fermentative effluents from hydrogen producing bioreactor as substrate for poly ( $\beta$ -OH) butyrate production with simultaneous treatment: an integrated approach. *Bioresour. Technol.* 101, 9382–9386.
- Mu, Y., Yu, H.Q., 2006. Biological hydrogen production in a UASB reactor with granules I: physicochemical characteristics of hydrogen-producing granules. *Biotechnol. Bioeng.* 94, 980–987.

- Mu, Y., Yu, H.Q., Wang, Y., 2006a. The role of pH in the fermentative H<sub>2</sub> production from an acidogenic granule-based reactor. *Chemosphere* 64, 350–358.
- Mu, Y., Zheng, X.J., Yu, H.Q., Zhu, R.F., 2006b. Biological hydrogen production by anaerobic sludge at various temperatures. *Int. J. Hydrogen Energy* 31, 780–785.
- Noike, T., Ko, I.B., Yokoyama, S., Kohno, Y., Li, Y.Y., 2005. Continuous hydrogen production from organic waste. *Water Sci. Technol.* 52, 145–151.
- Oh, S.E., Lyster, P., Bruns, M.A., Logan, B.E., 2004. Biological hydrogen production using a membrane bioreactor. *Biotechnol. Bioeng.* 87, 199–227.
- Peintner, C., Zeidan, A.A., Schnitzhofer, W., 2010. Bioreactor systems for thermophilic fermentative hydrogen production: evaluation and comparison of appropriate systems. *J. Cleaner production* 18, S15–S22.
- Ren, N., Wang, B., Huang, J.C., 1997. Ethanol-type fermentation from carbohydrate in high rate acidogenic reactor. *Biotechnol. Bioeng.* 54, 428–433.
- Ruzicka, M., 2006. The effect of hydrogen on acidogenic glucose cleavage. *Water Res.* 30, 2447–2451.
- Salerno, M.B., Park, W., Zuo, Y., Logan, B., 2006. Inhibition of biohydrogen production by ammonia. *Water Res.* 40, 1167–1172.
- Seghezzo, L., Zeeman, G., van Lier, J.B., Hamelers, H.V.M., Lettinga, G., 1998. A review: the anaerobic treatment of sewage in UASB and EGSB reactors. *Bioresource Technol.* 65, 175–190.
- Schmidt, J.E., Ahring, B.K., 1996. Granular sludge formation in upflow anaerobic sludge blanket (UASB) reactors. *Biotechnol. Bioeng.* 49, 229–246.
- Shen, L., Bagleya, D.M., Liss, S.N., 2009. Effect of organic loading rate on fermentative hydrogen production from continuous stirred tank and membrane bioreactors. *Int. J. Hydrogen Energy* 34, 3689–3696.
- Shen, L., Zhou, Y., Mahendran, B., Bagleya, D.M., Liss, S.N., 2010. Membrane fouling in a fermentative hydrogen producing membrane bioreactor at different organic loading rates. *J. Membrane Sci.* 360, 226–233.
- Shida, G.M., Barrosb, A.R., Marques, C., Cavalcante, E.L., Damjanovic, M.H.R.Z., Silvab, E.L., 2009. Long-term stability of hydrogen and organic acids production in an anaerobic fluidized-bed reactor using heat treated anaerobic sludge inoculum. *Int. J. Hydrogen Energy* 34, 3679–3688.
- Shin, H.S., Youn, J.H., 2005. Conversion of food waste into hydrogen by thermophilic acidogenesis. *Biodegradation* 16, 33–44.
- Shin, H.S., Youn, J.H., Kim, S.H., 2004. Hydrogen production from food waste in anaerobic mesophilic and thermophilic acidogenesis. *Int. J. Hydrogen Energy* 29, 1355–1363.
- Sung, S., Dague, R.R., 1992. Fundamental principles of the anaerobic sequencing batch reactor. In proceeding of the 47th Ind. Waste conference at Purdue Univ, West Lafayette.
- Ueno, Y., Fukui, H., Goto, M., 2007. Operation of a two-stage fermentation process producing hydrogen and methane from organic waste. *Environ. Sci. Technol.* 41, 1413–1419.
- Ueno, Y., Otsuka, S., Morimoto, M., 1996. Hydrogen production from industrial wastewater by anaerobic microflora in chemostat culture. *J. Ferment. Bioeng.* 82, 194–197.
- Vallero, M.V.G., Lettinga, G., Lens, P.N.L., 2005. Assessment of compatible solutes to overcome salinity stress in thermophilic (55° C) methanol-fed sulfate reducing granular sludge. *Water Sci. Technol.* 48, 195–202.
- Venetsaneas, N., Antonopoulou, G., Stamatiellatou, K., Kornaros, M., Lyberatos, G., 2009. Using cheese whey for hydrogen and methane generation in a two-stage continuous process with alternative pH controlling approaches. *Bioresource Technol.* 100, 3713–3717.
- Wang, C.H., Lin, P.J., Chang, J.S., 2006. Fermentative conversion of sucrose and pineapple waste into hydrogen gas in phosphate-buffered culture seeded with municipal sewage sludge. *Process Biochem.* 41, 1353–1358.
- Wang, Y., Mu, Y., Yu, H.Q., 2007. Comparative performance of two upflow anaerobic biohydrogen-producing reactors seeded with different sludges. *Int. J. Hydrogen Energy* 32, 1086–1094.
- Wu, S.Y., Hung, C.H., Lin, C.N., Chen, H., Lee, A.S., Chang, J.S., 2005. Fermentative hydrogen production and bacterial community structure in high-rate anaerobic bioreactors containing silicone-immobilized and self-flocculated sludge. *Biotechnol. Bioeng.* 93, 934–946.
- Wu, S.Y., Lin, C.N., Chang, J.S., 2003. Hydrogen production with immobilized sewage sludge in three-phase fluidized-bed bioreactors. *Biotechnol. Prog.* 19, 828–832.
- Yang, H., Shao, P., Lu, T., Shen, J., Wang, D., Xu, Z., Yuan, X., 2006. Continuous biohydrogen production from citric acid wastewater via facultative anaerobic bacteria. *Int. J. Hydrogen Energy* 31, 1306–1313.
- Yokoyama, H., Waki, M., Moriya, N., Yasuda, T., Tanaka, Y., Haga, K., 2007. Effect of fermentation temperature on hydrogen production from cow waste slurry by using anaerobic microflora within the slurry. *Appl. Microbiol. Biotechnol.* 74, 474–483.
- Yu, H., Zhu, Z., Hu, W., Zhang, H., 2002. Hydrogen production from rice winery wastewater in an upflow anaerobic reactor using mixed anaerobic cultures. *Int. J. Hydrogen Energy* 27, 1359–1365.
- Yu, H.Q., Mu, Y., 2006. Biological hydrogen production in a UASB reactor with granules II: reactor performance in 3-year operation. *Biotechnol. Bioeng.* 94, 988–995.
- Zhang, Z.P., Show, K.Y., Tay, J.H., David, T.L., Lee, D.J., 2008. Biohydrogen production with anaerobic fluidized bed reactors – a comparison of biofilm-based and granule-based systems. *Int. J. Hydrogen Energy* 33, 1559–1564.
- Zhang, Y., Shen, J., 2006. Effect of temperature and iron concentration on the growth and hydrogen production of mixed microflora. *Int. J. Hydrogen Energy* 31, 441–446.
- Zhang, Z.P., Show, K.Y., Tay, J.H., Liang, D.T., Lee, D.J., Jiang, W.J., 2006. Effect of hydraulic retention time on biohydrogen production and anaerobic microbial community. *Process Biochem.* 41, 2118–2123.
- Zhang, Z.P., Show, K.Y., Tay, J.H., Liang, D.T., Lee, D.J., Jiang, W.J., 2007. Rapid formation of hydrogen-producing granules in an anaerobic continuous stirred tank reactor induced by acid incubation. *Biotechnol. Bioeng.* 96, 1040–1050.